

6 Junie / June 2002
Maks Punte / Max Marks: 60
Tyd / Time: 120 min

Punte Marks	Intern Internal	Ekstern External

WTW 158 : CALCULUS

EKSAMEN / EXAMINATION

Eksterne eksaminator / External examiner: Dr NFJ van Rensburg

Van / Surname:

Voorname / First names:

Studentno:

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 Handtekening / Signature:

Studierigting/Field of Study:

Telefoonnommer gedurende eksamenperiode / Telephone number during exam period	
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Interne eksaminatore / Internal examiners	Interne eksaminatore / Internal examiners
Mev / Mrs A E du Preez	Mev / Mrs L Mostert
Mev / Mrs KH Jordaan	Prof MJ Schoeman

Lees eers die volgende instruksies

1. Slegs nie-programmeerbare sakrekenaars mag gebruik word.
2. Die vraestel bestaan uit bladsye 1 tot 10 (vrae 1-15). Kontroleer of u vraestel volledig is.
3. Doen alle krapwerk op die blanko bladsy(e). Dit word nie nagesien nie.
4. As u meer as die beskikbare ruimte vir 'n antwoord nodig het, gebruik dan ook die blanko bladsy(e) en dui dit duidelik aan deur 'n raam daarom te trek.
5. Geen potloodwerk word nagesien nie.
6. Korreksievloeistof (bv. Tipp-Ex) mag nie gebruik word nie.
7. Geen antwoordboek mag uit die lokaal geneem word nie.
8. Toon alle bewerkings duidelik

First read the following instructions

1. Only non-programmable calculators may be used.
2. The question paper consists of pages 1 to 10 (questions 1-15). Check if your paper is complete.
3. Scribbling can be done on the empty page(s). This will not be marked.
4. If you need more than the available space for an answer, you may also use the empty page(s). Indicate it clearly by framing it.
5. Work in pencil will not be marked.
6. Use of correction fluid (e.g. Tipp-Ex) is not allowed.
7. No test scripts may be removed from the venues.
9. Show all computations clearly

VRAAG 1 (TEGNIEKE) / QUESTION 1 (TECHNIQUES)

(1.1) Vind y as $\frac{dy}{dx} = \frac{1}{x}$, $y(1) = 2$.

Find y if $\frac{dy}{dx} = \frac{1}{x}$, $y(1) = 2$.

[1]

(1.2) Bepaal / Find

(1.2.1) $\int x^3 \sqrt{x} \, dx$

[1]

(1.2.2) $\int e^{5x} \, dx$

[1]

(1.2.3) $\int \sinh 4x \, dx$

[1]

(1.2.4) $\int x e^{x^2} \, dx$.

[1]

(1.3) Bepaal die volgende limiete indien hulle bestaan.

Evaluate the following limits if they exist.

(1.3.1) $\lim_{x \rightarrow \infty} \frac{x^2 + x + 1}{x^3 + 2x^2 + 1}$

[1]

$$(1.3.2) \lim_{x \rightarrow 1^+} \frac{x^2 - 3x + 2}{x^2 - 2x + 1}.$$

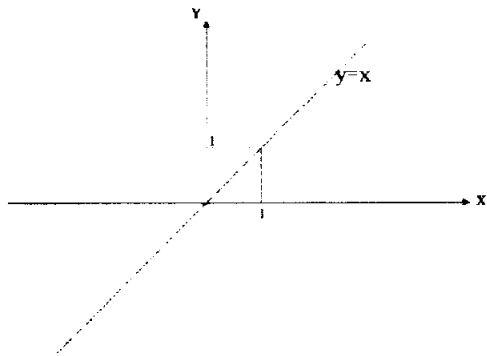
[1]

(1.4) Laat / Let $f(x) = \sin x$ en / and $g(x) = \sin^{-1} x$.

[Notasie / Notation: $\sin^{-1} x = \arcsin x = \text{bgsin } x$]

Skets die grafieke van f en g op dieselfde assestelsel.

Sketch the graphs of f and g on the same set of axes.



[2]

VRAAG 2 / QUESTION 2

Vind die afgeleide van f .

Find the derivative of f .

$$(2.1) f(x) = \sqrt{x^2 + 1} \sinh x.$$

[2]

$$(2.2) f(x) = \frac{1}{x} + \tan 3x.$$

[2]

$$(2.3) f(x) = \tan^{-1}(3x + 1) \text{ [Notasie / Notation : } \tan^{-1} x = \arctan x = \text{bgtan } x].$$

[1]

VRAAG 3 / QUESTION 3

Bepaal / Determine

(3.1) $\int \sqrt{\frac{5}{x}} dx$

[1]

(3.2) $\int e^x \sqrt{1+e^x} dx$

[2]

(3.3) $\int \frac{x+1}{x^2+2x} dx.$

[2]

(3.4) $\int \sin^2 5x dx.$

[3]

VRAAG 4 / QUESTION 4

Stel vas of $y = 2 \sin x + 3 \cos x$ die vergelyking $\frac{d^2 y}{dx^2} + y = 0$ bevredig.

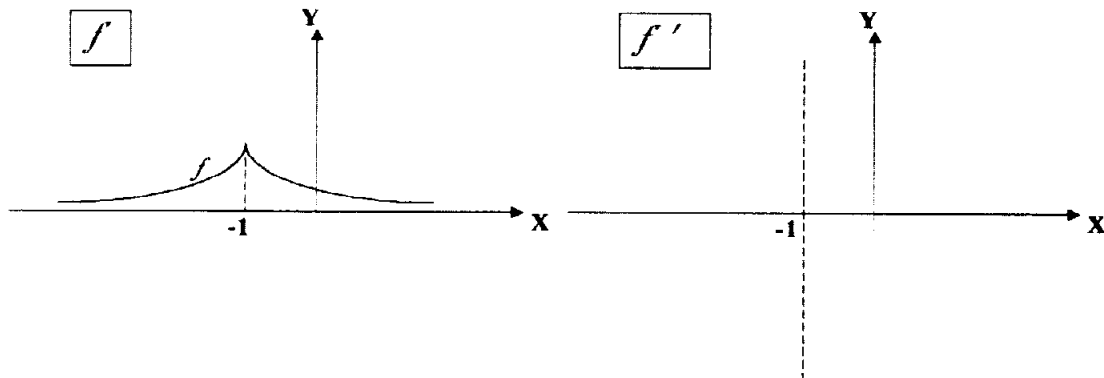
Determine whether $y = 2 \sin x + 3 \cos x$ satisfies the equation $\frac{d^2 y}{dx^2} + y = 0$.

[2]

VRAAG 5 / QUESTION 5

Die grafiek van f is gegee. Skets 'n grafiek van f' (die afgeleide van f).

The graph of f is given. Sketch a graph of f' (the derivative of f).

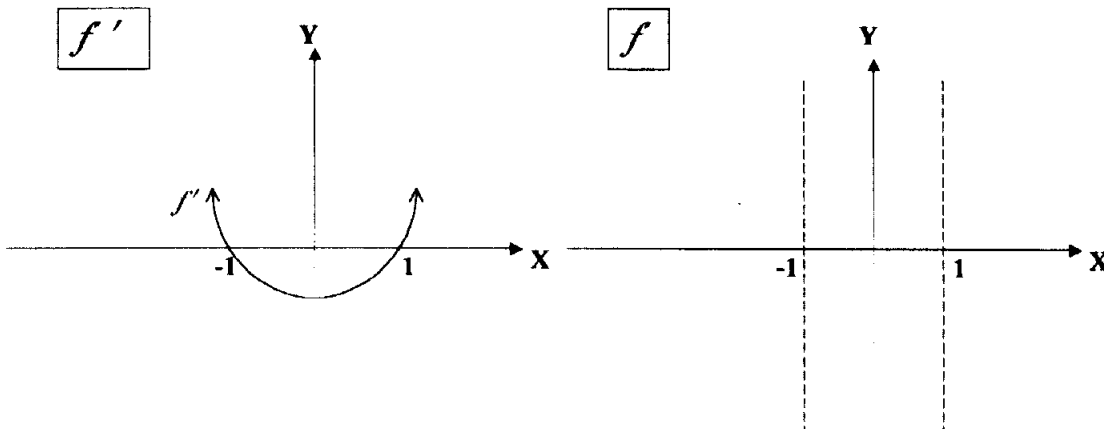


[1]

VRAAG 6 / QUESTION 6

Die grafiek van f' (die afgeleide van f) is gegee. Skets 'n moontlike grafiek van f .

The graph of f' (the derivative of f) is given. Sketch a possible graph of f .



[1]

VRAAG 7 / QUESTION 7

Gebruik implisiete differensiasie om die vergelyking van die raaklyn aan $x \cos y = 1$ in die punt $(2, \frac{\pi}{3})$ te kry.

Use implicit differentiation to find the equation of the tangent line to $x \cos y = 1$ at the point $(2, \frac{\pi}{3})$.

[3]

VRAAG 8 / QUESTION 8

Is die volgende bewerings waar of onwaar. Indien waar, verduidelik jou antwoord, indien onwaar, gee 'n teenvoorbeeld.

Are the following statements true or false. If true, explain your answer, if false give a counterexample.

(8.1) Laat / Let $f(x) = \frac{1}{x-3}$.

Daar bestaan 'n $c \in (0, 6)$ sodat $f'(c) = \frac{f(6) - f(0)}{6}$.

There exists a $c \in (0, 6)$ such that $f'(c) = \frac{f(6) - f(0)}{6}$.

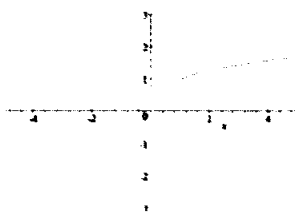
[1]

(8.2) Gegee die grafiek van $f(x) = x^{1/3}$ soos dit deur rekenaar sagteware voorgestel word.

Die skets is korrek want die rekenaar maak nie foute nie.

Given the graph of $f(x) = x^{1/3}$ as it is presented by computer software.

The sketch is correct, since the computer never makes a mistake.



Scientific Workplace



Matlab

[1]

(8.3) As 'n funksie diskontinu in $x = a$ is, dan is die funksie nie differensieerbaar in $x = a$ nie.

If a function is discontinuous at $x = a$ then it is not differentiable at $x = a$.

[1]

VRAAG 9 / QUESTION 9

Op 'n sekere dag word die aantal voertuie wat verby 'n punt beweeg, gegee deur $F = \frac{v}{90 + 0.02v^2}$

met v die snelheid van die voertuie. Watter snelheid sal die verkeersvloei maksimeer?

On a certain day the number of vehicles passing a specific spot, is given by $F = \frac{v}{90 + 0.02v^2}$

with v the velocity of the vehicles. Which velocity will maximize the traffic flow?

[3]

VRAAG 10 / QUESTION 10

Gegee / Given $f(x) = \begin{cases} \sqrt{x} & , x < 0 \\ 3 - x & , 0 \leq x < 3 \\ (x - 3)^2 & , x > 3 \end{cases}$

Gee (met redes) al die punte waar f diskontinu is.

Give (with reasons) all the points where f is discontinuous.

[2]

VRAAG 11 / QUESTION 11

Laat / Let $f(x) = \frac{x}{(x-1)^2}$, $f'(x) = \frac{-(x+1)}{(x-1)^3}$ en / and $f''(x) = \frac{2x+4}{(x-1)^4}$.

(11.1) Vind die kritieke getal(le) van f .

Find the critical number(s) of f .

[2]

(11.2) Vind die lokale (relatiewe) ekstreme van f en spesifiseer dit.

Find the local (relative) extremes of f and specify it.

[2]

(11.3) Vind die horisontale asimptote van f .

Find the horizontal asymptotes of f .

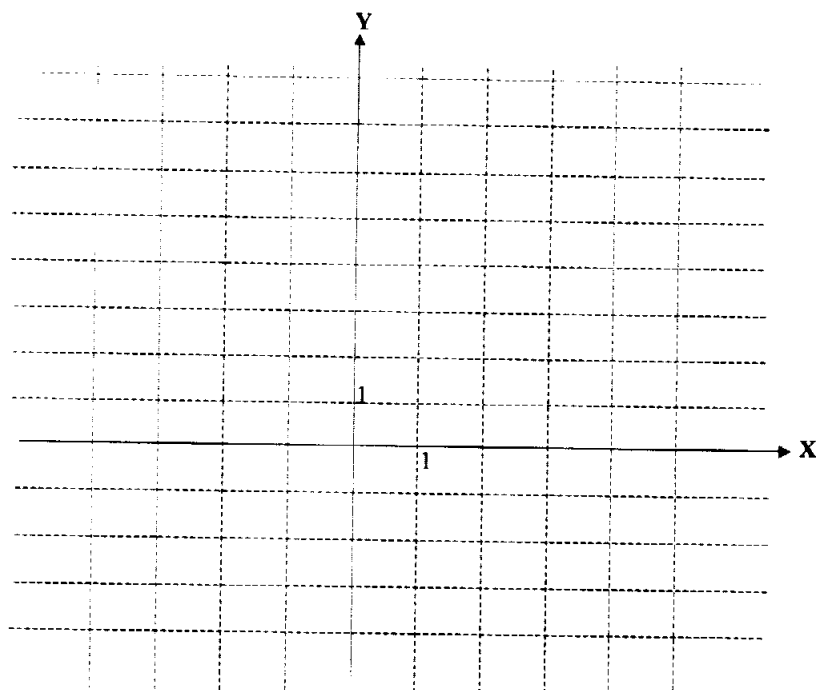
[2]

(11.4) Vind die interval(le) waar f konkaaf na bo is.

Find the interval(s) where f is concave upwards.

[1]

(11.5) Skets f .
Sketch f .



[3]

VRAAG 12 / QUESTION 12

Skets 'n funksie f wat aan al die volgende voorwaardes voldoen:

Sketch a function f that satisfies all the following conditions:

• $f(2) = f(4) = 0$

• $f(2) = f(4) = 0$

• $f'(x) > 0$ as $0 < x < 3$

• $f'(x) > 0$ if $0 < x < 3$

• $f'(3)$ bestaan nie

• $f'(3)$ does not exist

• $f'(x) < 0$ as $x > 3$

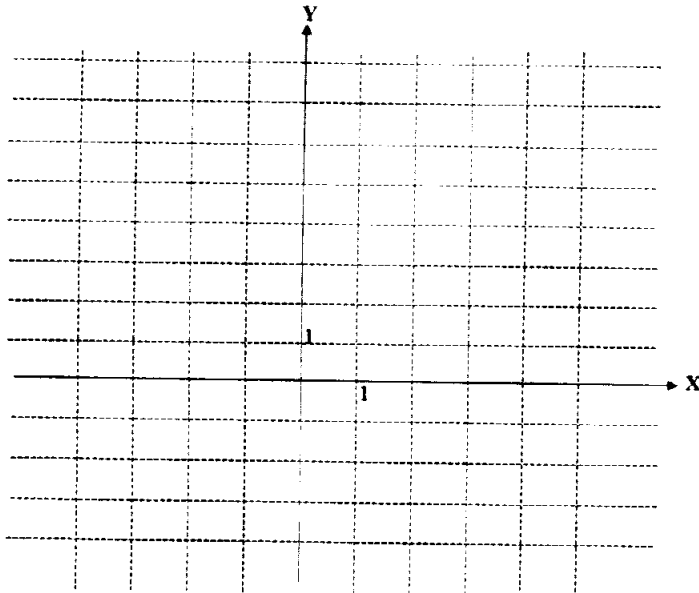
• $f'(x) < 0$ if $x > 3$

• $f''(x) > 0$ as $x \neq 3$

• $f''(x) > 0$ if $x \neq 3$

• f is ewe.

• f is even.

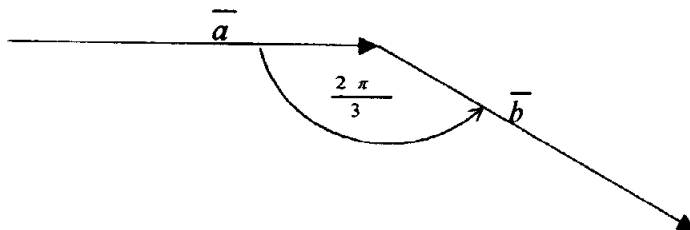


[3]

VRAAG 13 / QUESTION 13

Gegee $|\vec{a}| = 2$ en $|\vec{b}| = 3$ met $\vec{a}$ en $\vec{b}$ soos in die skets. Bereken $\vec{a} \cdot \vec{b}$.

Given $|\vec{a}| = 2$ and $|\vec{b}| = 3$ with $\vec{a}$ and $\vec{b}$ as in the sketch. Evaluate $\vec{a} \cdot \vec{b}$.



[3]

VRAAG 14 / QUESTION 14

Bepaal 'n vergelyking van die vlak deur die punte $A(1, 2, -3)$, $B(2, 3, 1)$ en $C(0, -2, -1)$.

Determine an equation of the plane through the points $A(1, 2, -3)$, $B(2, 3, 1)$ and $C(0, -2, -1)$.

[4]

VRAAG 15 / QUESTION 15.

Gegee die parametrisiese vergelykings van die lyne ℓ_1 en ℓ_2 .

met $\ell_1 : x = 4t + 2, y = 3, z = -t + 1$

en $\ell_2 : x = 2s + 2, y = 2s + 3, z = s + 1$.

Stel vas of die lyne ℓ_1 en ℓ_2 mekaar sny. Indien wel, vind die snypunt van die lyne.

Given the parametric representation of ℓ_1 and ℓ_2

with $\ell_1 : x = 4t + 2, y = 3, z = -t + 1$

and $\ell_2 : x = 2s + 2, y = 2s + 3, z = s + 1$.

Determine whether the lines ℓ_1 and ℓ_2 intersect. If they do, find the point of intersection.

[3]